

## 2021-2025 Sports Injury s Medical Operational Audit

### Situation

This report analyses 15,000 medical records and the management of a medical budget of **€27.7M** over the course of 4 seasons between 2021 and 2025 to evaluate the financial efficiency and clinical effectiveness of the organization's injury management. Key findings include a stable spend to workload ratio of roughly €1,360 for every 1,000 minutes played and a modest, statistically non-significant rise in average recovery time in the 3rd season (2023-2024).

### Task

My objective was to perform an audit on:

- Medical efficiency by measuring the days till recovery against the cost of treatment.
- Injury data for age and gender to determine if gender-specific safety protocols are required.
- Treatment method effectiveness in restoring athletes to full health.
- Which coaching styles were getting injured.
- What regions were getting the greatest number of injuries.
- What playing surfaces caused the most injuries.
- Average recovery time.

### Action

I engineered a **Star Schema** data model and developed advanced **DAX measures** to normalize the data.

Specifically, I:

- Established a multi dimensional framework to measure days to recovery against the cost of treatment across the different injury events.
- Compared treatment methods by full-recovery rate and average days to recovery, segmented by severity to ensure a fair comparison.
- Developed a custom Spend Efficiency metric to show the cost per 1,000 Minutes Played to neutralize the impact of fluctuating season lengths and athlete exposure.
- Analysed coach and team performance metrics which revealed high cardinality in the coaching dataset with an almost 1:1 injury-to-coach ratio.
- Conducted a regional risk assessment across four European regions.
- Cross examined 14 unique playing surfaces against sports and competition tiers to compare where injuries were recorded.
- Conducted an injury volume analysis between Professional and Amateur athlete tiers to identify the primary drivers of clinical workload.
- Dug deeper into the increase in the average recovery time for the 2023-2024 season even though the total spent for that year was lower than the previous year.
- Analysed how age and gender affect injury occurrence.

## Results

- Rest had the highest full recovery rate of 82.3%, followed by medication at 81.2%, ice/heat therapy at 78.6% and massage 78.4%. Rest led at every severity level. Average days to recovery did not differ significantly between treatments.
- I showed that despite a swing of about €4.3M in spending between seasons, spend per 1,000 minutes played stayed between €1,312 and €1,390 (about €1,364 overall).
- I discovered that 99% of coach entries contained only a single injury event, rendering an individual coach's consistency metrics statistically insignificant.
- Central Europe reported the highest injury volume with 33.7% of cases. Severe cases made up 14.0% of injuries in Central Europe, 15.1% in Western, 15.4% in Southern and 16.0% in Northern Europe. These differences are small and not statistically significant.
- Indoor Courts and Grass emerged as the highest injury causing environments. However, the data revealed a consistent 65-70% increase in amateur injuries when compared to professionals across all surfaces.
- There was significant amateur strain, where non-professional athletes account for 70.7% (10,610 cases) of the total injury volume and professionals having 29.3% (4,390 cases).
- The 2023-2024 season exhibited the highest recovery time of 51 days with further analysis indicating it was a period of more severe injuries that cost the team despite having less total injuries. This is also supported by the spend efficiency being the highest in that season.
- Older players are more likely to get injured with athletes over 30 having the most injuries compared to those below 20. Injuries are spread evenly across ages 16-40 (about 20% per five-year band). While women account for 65.5% of records, the data doesn't tell us how many players are women, therefore, this cannot be read as higher risk.

## Recommendations

1. Track injuries per hours played by playing surface within each sport before investing in surface upgrades.
2. Collect exposure data (athlete headcount and hours played by tier) so injury rates per 1,000 hours can be compared before investing in tier-specific programmes.
3. Enhance preventive care and monitoring: Introduce regular screenings, physiotherapy check-ins, and load management strategies to reduce recurrence rates (currently 19.83%).
4. Regional monitoring: Track injury severity by region over time. Current differences (14.0-16.0% severe) are too small to justify region-specific programmes.
5. Gender and age-specific protocols: Record roster size and exposure by age and gender before designing age- or gender-specific protocols.